

EDUCATION

MS in Computer Science University of Oregon, Eugene, OR <i>Achievements:</i> Lokey Award for Graduate Excellence (2024-2025) Frank Vignola Microgrant (2025) – Awarded \$1,500	Sep 2024 – Mar 2027
B.Tech in Computer Science and Engineering Vellore Institute of Technology, TN, India <i>Achievements:</i> Awarded Highest Academic Scholarship (2019–2023, Indian Embassy) Vice President, 5th Pillar NGO	May 2019 – Apr 2023

TECHNICAL SKILLS

Systems/Web3: Distributed Systems, Blockchain State Simulation, Verifiable Storage, Authenticated Data Structures
Cryptography: Cryptographic Proofs, Commitment Schemes, Merkle Trees, Integrity Verification, Zero-Knowledge Proofs
Languages: Java, Go, Python, JavaScript, TypeScript, HTML/CSS
Backend: Node.js/Express, REST APIs, GraphQL, API Design
Frontend: React.js, Redux, SASS, Polymer.js, LitElement
Databases: MySQL, MongoDB, SQL Query Optimization
DevOps & Tools: Git/GitHub, GitLab, Docker, Postman, JIRA, JavaScript Testing

PROFESSIONAL AND RESEARCH EXPERIENCE

Leapfrog Technology Inc., Kathmandu, Nepal Software Engineer	Apr 2023 – June 2024
- Designed and implemented backend APIs using Micronaut, supporting scalable and maintainable service architecture. - Optimized RESTful APIs, reducing response times by 30% through endpoint refactoring, query tuning, and performance profiling. - Developed responsive React.js interfaces with reusable components, improving UI maintainability and cross-browser consistency. - Integrated REST APIs with React front ends, ensuring reliable data flow, input validation, and clear error handling. - Collaborated with product, design, and engineering teams in Agile development cycles to ship production-facing features. - Participated in debugging, testing, code review, and iterative development to improve correctness, maintainability, and delivery speed. - Streamlined engineering workflows, reducing project timelines by 20% through improved development processes and Agile practices. - Led a team of annotators, improving data-labeling accuracy by 25% and reducing false positives in downstream machine-learning workflows.	

University of Oregon, Eugene, OR Graduate Researcher	Sep 2024 - Present
- Research secure and reliable distributed systems, with a focus on verifiable blockchain history and metastability in overloaded services. - Design authenticated data structures for historical blockchain queries, enabling users to verify past states without trusting centralized providers. - Built a custom Ethereum-style state simulator in Go to evaluate proof-generation latency, verification cost, and storage overhead. - Developed experiments for metastable failure scenarios, measuring queue growth, timeout behavior, latency spikes, and recovery under capacity degradation. - Study reliability mechanisms for distributed services, including early distress signaling, helper activation, and fresh-work redirection during overload.	

Teaching Assistant
- Delivered web programming lectures and hands-on debugging labs for 140+ students, strengthening practical coding and problem-solving skills. - Led lab sessions, evaluated assignments, and provided individualized feedback to improve students’ programming and debugging proficiency.

PROJECTS

- Smaran: Serving Authenticated Time-Travel Queries — 2025:** Lead development of Smaran, a decentralized proof system that makes **blockchain** history verifiable without trusting centralized providers. Design authenticated data structures for historical range queries, integrate cryptographic proof generation with blockchain-style workflows, and build Go-based simulation/benchmarking components to evaluate proof-generation latency, verification cost, and storage overhead.
- Cry-for-Help: Reliability Support for Metastable Failures — 2025–Present:** Research system support for detecting and mitigating metastable failures in distributed services before overload and retry loops become self-sustaining. Design an in-network coordination protocol where distressed servers request help, redirect fresh work, and activate helper nodes over shared state; build experiments to measure queue growth, timeout behavior, latency spikes, and recovery under capacity degradation.
- Aspen Biosciences — 2023:** Built features for a drug-discovery management platform used by scientists to manage experiments, chemistry workflows, and inventory. Integrated project-management, experiment-tracking, and inventory-management functionality into a unified web application to improve research workflow efficiency.
- Vyaguta ERP System — 2023:** Developed backend APIs and responsive dashboard interfaces for employee and project-management workflows. Optimized SQL queries and REST API endpoints, improving dashboard performance, data access reliability, and operational efficiency across multiple internal workflows.
- VIT Connects:** Designed a cross-platform application using Flutter and Firebase to support course collaboration, task management, and student-faculty interaction, and applied HCI principles through user personas and usability testing to improve engagement and user experience.